

Orbit-Resolved Cumulant Pressure for a Minimal Hénon Reflection Defect

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Abstract

A canonical Euler product for odd reflection packets can assign one aggregate mean to every period, but this loses the distribution of individual orbit weights. We compute that loss exactly for the minimal cross-axis observable $\chi_R(s) = \mathbf{1}\{s_{-1} = s_1\}$ in the full Hénon horseshoe. Odd cyclic decimation conjugates its orbit sum to nearest-neighbor equality energy on a reflected binary chain. For $n = 2m + 1$, the complete packet polynomial is

$$F_n(q) = 2q(1 + q^2)^m,$$

and the primitive polynomial satisfies the dilation Möbius law $E_n(q) = \sum_{k|n} \mu(k) F_{n/k}(q^k)$. Consequently the orbit-resolved primitive pressure is

$$P_{\text{orb}}(s) = \frac{1}{2} \log(1 + e^{-2s}).$$

The aggregate-mean pressure is $\frac{1}{2} \log 2 - \frac{s}{2}$, so their exact gap is $\frac{1}{2} \log \cosh s$, strictly positive for every $s \neq 0$. This proves a global cumulant obstruction to mean-field replacement while supplying exact coefficient data for a full orbit-resolved Euler product. It gives no arithmetic trace or operator.

1 The missing distribution

For the area-preserving map $H(x, y) = (6 - x^2 - y, x)$, the real horseshoe is conjugate to the full two-shift and the reversor acts as sequence reversal [2, 1]; see Lind and Marcus [3] for the symbolic framework. At odd period, write A_n for the primitive marked reflection packet. The canonical packet-mean Euler product assigns to all elements of A_n the common weight generated by their average. That construction is intrinsic under coboundaries, but it does not retain within-period fluctuations.

We use the minimal cross-axis separator

$$\chi_R(s) = \mathbf{1}\{s_{-1} = s_1\}, \quad S_n \chi_R(s) = \sum_{j=0}^{n-1} \chi_R(\sigma^j s). \quad (1)$$

The complete and primitive moment polynomials are

$$F_n(q) = \sum_{\substack{s: \sigma^n s = s \\ s_{-j} = s_j}} q^{S_n \chi_R(s)}, \quad (2)$$

$$E_n(q) = \sum_{s \in A_n} q^{S_n \chi_R(s)}. \quad (3)$$

2 A reflected rank-two transfer matrix

Proposition 2.1 (Odd decimation). *Let $n = 2m + 1$. A reflection-fixed word is determined by $a_j = s_j$, $0 \leq j \leq m$, and*

$$S_n \chi_R = 1 + 2 \sum_{j=0}^{m-1} \mathbf{1}\{a_j = a_{j+1}\}. \quad (4)$$

Consequently

$$F_{2m+1}(q) = 2q(1 + q^2)^m. \quad (5)$$

Proof. Multiplication by 2 permutes $\mathbb{Z}/n\mathbb{Z}$, commutes with $j \mapsto -j$, and sends the distance-two comparisons in (1) to nearest-neighbor comparisons. Under reflection, each path edge occurs twice and the edge joining the two terminal reflected sites is always equal. This gives (4). Equivalently, with

$$T(q) = \begin{pmatrix} q^2 & 1 \\ 1 & q^2 \end{pmatrix},$$

one has $F_{2m+1}(q) = q(1, 1)T(q)^m(1, 1)^\top$. The vector $(1, 1)$ has eigenvalue $1 + q^2$, proving (5). $\square$

Remark 2.2 (Exact energy distribution). *There are $2\binom{m}{k}$ reflected words of energy $1 + 2k$. Thus the equality increments form an exact binomial ledger; no asymptotic independence is assumed.*

3 Primitive subtraction with energy dilation

Ordinary Möbius subtraction would be wrong because repetition multiplies both length and energy.

Theorem 3.1 (Primitive moment polynomial). *For odd n ,*

$$F_n(q) = \sum_{d|n} E_d(q^{n/d}), \quad (6)$$

$$E_n(q) = \sum_{k|n} \mu(k) F_{n/k}(q^k). \quad (7)$$

Proof. A word of least period $d \mid n$ repeats n/d times, so its n -step energy is $(n/d)S_d \chi_R$. Partitioning (2) by least period gives (6); Möbius inversion in the semigroup of the dilations $q \mapsto q^k$ gives (7). $\square$

At $q = 1$, (7) recovers the primitive count $D_n = \sum_{d|n} \mu(n/d)2^{(d+1)/2}$. Coefficientwise nonnegativity is also immediate from the defining sum (3).

4 Nonlinear pressure and the exact gap

Theorem 4.1 (Orbit-resolved pressure). *For each fixed real s ,*

$$P_{\text{orb}}(s) := \lim_{\substack{n \rightarrow \infty \\ n \text{ odd}}} \frac{1}{n} \log E_n(e^{-s}) = \frac{1}{2} \log(1 + e^{-2s}). \quad (8)$$

Proof. The $k = 1$ term in (7), using (5), has the displayed rate. A proper divisor has odd $k \geq 3$ and rate

$$\frac{1}{2k} \log(1 + e^{-2ks}) < \frac{1}{2} \log(1 + e^{-2s}).$$

For $s \geq 0$ this follows from monotonicity. For $s < 0$, put $x = e^{-2s} > 1$ and use $1 + x^k < (1 + x)^k$. Proper-divisor terms are therefore exponentially smaller; the divisor count is subexponential. Since $E_n(e^{-s})$ is a positive defining sum, the leading asymptotic follows. $\square$

The fair Bernoulli expectation of χ_R is $1/2$. The aggregate-mean packet pressure is consequently

$$P_{\text{mf}}(s) = \frac{1}{2} \log 2 - \frac{s}{2}. \quad (9)$$

Corollary 4.2 (Exact cumulant obstruction). *For every real s ,*

$$P_{\text{orb}}(s) - P_{\text{mf}}(s) = \frac{1}{2} \log \cosh s. \quad (10)$$

The gap is zero only at $s = 0$, and $P''_{\text{orb}}(0) = 1/2$.

Proof. Use $1 + e^{-2s} = 2e^{-s} \cosh s$ in (8). Strictness and the second derivative are immediate. $\square$

This is not merely a limiting Jensen inequality. For every odd $n \geq 5$, the primitive reflected word with one centered 1 has energy $n - 2$, while the centered block 111 has energy $n - 4$. Thus finite primitive variance is already positive.

5 Audit and route boundary

The certificate constructs exact integer polynomials through $n = 31$. An independent implementation enumerates every reflected word through $n = 17$ and builds the binomial rows by Pascal recurrence. Eight tests pass in normal and optimized modes, six dependencies are hash locked, and 25 claim mutations are rejected.

The result upgrades the packet pressure from a first moment to a full cumulant generating function for one natural potential. It does not attach rational primes, von Mangoldt amplitudes, a Fredholm operator, or a self-adjoint spectrum. Route A remains exploratory and Route B is not authorized. The exact polynomials (7) are now the natural input for a genuinely orbit-resolved Euler product and its analytic boundary.

References

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